

CSS (Cascading Style Sheet)

Animation :-



- An animation makes an element change gradually from one style to another.
- You can also specify the changes in percentage.
- 0% specify the start of the animation and 100% specify its completion.

```
<style>
div
{
  width: 150px;
  height: 150px;
  background: red;
  position: relative;
  -webkit-animation: myfirstanimation 5s;
}

@-webkit-keyframes myfirstanimation
{
  0% {background:red; left:0px; top:0px;}
  25% {background:yellow; left:300px; top:0px;}
  50% {background:blue; left:200px; top:300px;}
  75% {background:green; left:0px; top:200px;}
  100% {background:red; left:0px; top:0px;}
}

</style>
```



<div></div>

Standard Syntax -

```
<style>
div
{
  width: 150px;
  height: 150px;
  background: red;
  position: relative;
  animation: myfirstanimation 5s;
}

@keyframes myfirstanimation
{
  0% {background:red; left:0px; top:0px;}
  25% {background:yellow; left:300px; top:0px;}
  50% {background:blue; left:300px; top:200px;}
  75% {background:green; left:0px; top:200px;}
  100% {background:red; left:0px; top:0px;}
}

</style>
```



<div></div>

Basics of CSS Animation :-

1. @keyframes - Defines the stages of the animation (what happens at each step).
2. animation property - Applies the animation to an element.

Common Animation Properties :-

- animation-name - Name of keyframes.
- animation-duration - Time (e.g., 2s, 500ms).
- animation-timing-function - linear, ease, ease-in, ease-out, ease-in-out.
- animation-delay - Start delay.
- animation-iteration-count - 1, 3, infinite.
- animation-direction - normal, reverse, alternate.
- animation-fill-mode - none, forwards, backwards, both.
- animation - Shorthand property (all in one line).

Example –

HTML Code –

```
CSS > <> animation.html > html
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4  |   <meta charset="UTF-8">
5  |   <meta name="viewport" content="width=device-width, initial-scale=1.0">
6  |   <title>Document</title>
7  |   <link rel="stylesheet" href="./animation.css">
8  </head>
9  <body>
10 |   <div></div>
11 </body>
12 </html>
```

CSS Code –

```

CSS > # animation.css > @keyframes xyz
1  div{
2      height: 300px;
3      width: 300px;
4      background-color: cadetblue;
5      animation: xyz 5s ease-in-out 3s infinite;
6  }
7
8  @keyframes xyz {
9      0%{
10         background: linear-gradient(blue, black);
11         border-radius: 20px;
12         transform: rotate(0deg);
13     }
14
15     20%{
16         background: linear-gradient(red, black);
17         border-radius: 40px;
18         transform: rotate(45deg);
19     }
20
21     40%{
22         background: linear-gradient(orange, black);
23         border-radius: 60px;
24         transform: rotate(90deg);
25         float: right;
26     }
27
28     60%{
29         background: linear-gradient(purple, white);
30         border-radius: 80px;
31         transform: rotate(180deg);
32         transform: scale(1.6);
33     }
34
35     80%{
36         background: linear-gradient(white, red);
37         border-radius: 100px;

```

CSS Transform :-

- transform is used to change the shape, size, and position of an element.
- It works in 2D and 3D space.
- It does not affect other elements around it (does not change layout flow).

Common transform functions are:

- `translate(x, y)` → moves element.
- `rotate(angle)` → rotates element.
- `scale(x, y)` → zooms in/out.
- `skew(x, y)` → slants element.
- `matrix()` → combines multiple transforms.

Example –

```
<!DOCTYPE html>

<html>

<head>

<style>

.box {

  width: 100px;

  height: 100px;

  background: coral;

  margin: 50px;

  transform: rotate(45deg) scale(1.2);

}

</style>

</head>

<body>


<h2>Transform Example</h2>

<div class="box"></div>


</body>

</html>
```

CSS Transition :-

- transition makes property changes smooth over time.
- It requires a trigger (like hover, focus, or JavaScript).

Key properties are:

- transition-property → which property to animate.
- transition-duration → how long animation takes.
- transition-timing-function → speed curve (ease, linear).
- transition-delay → when animation should start.

Example –

```
<!DOCTYPE html>
<html>
<head>
<style>
button {
  padding: 15px 30px;
  font-size: 18px;
  background: royalblue;
  color: white;
  border: none;
  border-radius: 8px;
  cursor: pointer;

  transition: background 0.5s ease,
  transform 0.5s ease;
}

button:hover {
  background: darkorange;
  transform: scale(1.2) rotate(10deg);
}
</style>
</head>
<body>

<h2>Transition Example</h2>
<button>Hover Me</button>
```

</body>

</html>